

Complex Composite Restoration (Principles of Preparation)

Composite resin restorations are now central to modern operative dentistry because they offer excellent aesthetics, allow for a more conservative approach, and can bond directly to the tooth structure. Unlike amalgam restorations, which require mechanical retention and thus removal of additional tooth structure, composites rely on adhesive bonding. This shift fundamentally changes cavity design principles. Therefore, it is no longer necessary to remove healthy tooth tissue solely to create mechanical retention, thereby preserving tooth strength and integrity.

A cavity is considered complex when it involves more than two surfaces, undermines one or more cusps, weakens the remaining tooth structure, or presents a high cavity configuration factor (C-factor). In literature, it is also mentioned a posterior cavity can be considered as complex because of three factors: firstly the depth of its proximal margins, secondly the structural vulnerability, and lastly because of its size.

Complex restorations often begin with teeth that show old amalgam restorations, secondary caries, marginal breakdown, or cuspal cracks. These teeth may present with hypersensitivity or structural compromise. The first decisive step is proper isolation using rubber dam. Isolation is not merely for moisture control; it improves visibility, prevents contamination, and allows predictable adhesive procedures. Without isolation, the reliability of bonding and long-term success decreases significantly.

Cavity preparation:

According to Sturdevant and Summitt, cavity preparation for composite restorations must satisfy the objectives (biological, mechanical, and aesthetic) while maximising the preservation of sound tooth structure. The outline form of the cavity is dictated by the extent of caries, the presence of defective restorations, the need for access and visibility, and the availability of sound enamel for bonding. There should be no extension for prevention; all carious and weak unsupported enamel should be removed, marginal ridges preserved whenever possible, and unnecessary cusp removal avoided, therefore, cusp/wall reinforcement would be recommended rather than reduction.

After removal of caries or old restoration, the clinician evaluates the remaining tooth structure. Caries excavation is performed conservatively, preserving affected but remineralizable dentin when appropriate. The cavity is then cleaned, often with antimicrobial agents such as chlorhexidine, to reduce bacterial load. This stage determines the final cavity geometry and the amount of cuspal support remaining.

The internal form of the cavity should feature rounded internal line angles and smooth, flowing walls, with flat pulpal and gingival floors where present. This design reduces stress concentration during polymerisation shrinkage, improves adaptation of the

composite resin, and minimises the risk of crack initiation in the tooth structure. Internal rounded line angle can decrease the C-factor since it reduces the number of walls i.e. When two surfaces are joined by a smoothly rounded line angle, they behave as a single wall, which reduces the impact of C-factors during polymerization. The C-factor is defined as the ratio of bonded to unbonded surfaces in a cavity. A higher C-factor leads to increased polymerisation shrinkage stress, which can compromise the restoration's integrity. In addition, it is essential to help the tooth and restoration withstand occlusal forces. Unsupported enamel must be removed, and thin or undermined cusps, especially those less than 2 mm thick for vital tooth, require reduction and coverage with composite. The recommended cusp reduction is between 1.5 and 2.0 mm. Therefore, when complex cavities involve weak cusps, coverage is indicated if the cusp is functional and under heavy occlusal load, or if it is cracked or weakened.

Retention in composite restorations is achieved primarily through micromechanical and chemical bonding, rather than mechanical undercuts. It is important not to create undercuts solely for retention, as this may weaken the tooth structure. Preserving enamel is critical for maximum bond strength, and the selection of an appropriate adhesive system is vital. However, in some cases, after caries removal it might be some undercuts or proximal boxes which add an extra retentive means against dislodgment.

Margin location and design are also critical. Enamel margins provide optimal bonding, and beveling may be used except in areas of heavy occlusal load or difficult to access. However, we should note that bevel should be created on enamel and not dentine. Dentin or cementum margins have lower bond strength.

Some teeth present large occlusal cavities with thin enamel cups but sound dentin. These cases are challenging because cusp strength is difficult to judge. Over-reduction to dentine to increase cavity depth may weaken the tooth unnecessarily, whereas application of thin restoration may lead to fracture. Objective structural evaluation and sound clinical judgment are essential in deciding whether cusp coverage is required, probably cusp coverage using indirect restoration is the recommended treatment to withstand occlusal force without unnecessary sound dentine removal from the cavity floor.

Preparation guide of a weak cusp:

However, when there is an extensive caries that requires increasing the width of cavity to remove all caries, there is a chance of weakening of remaining cusp or wall, and in this case there is a rule for cusp reduction and coverage according to **Sturdevant and Summitt**; If extension from a primary groove toward the cusp tip is no more than half the distance, no cusp reduction and coverage should be done; if the extension is one half to two thirds of the distance, consider cusp reduction and coverage; if the extension is more than two thirds of the distance, usually reduce the cusp and cover it with restorative material. Cusp reduction and coverage has also been referred to as “cusp capping.” (Figure 1 & 2).

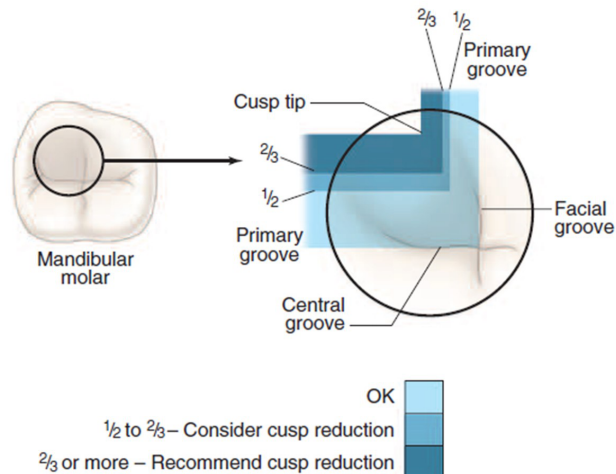


Figure 1: decision of cusp reduction by **Sturdevant and Summitt**.

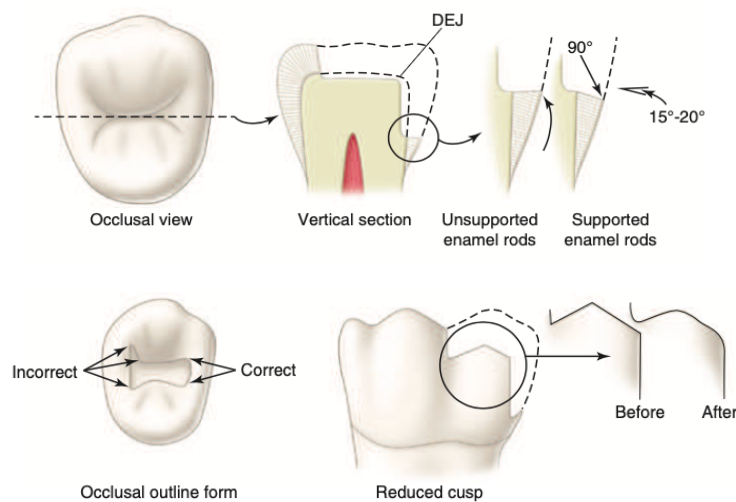


Figure 2: Vertical section of Class II tooth preparation. Gingival floor enamel (and margin) is unsupported by dentin and friable unless removed. The junctions of enamel walls (and respective margins) should be slightly rounded, whether obtuse or acute.

Cusp reduction:

This is an example of complex cavity preparation step by step with a reduction of a weak cusp (Figure 3):

Orange: 1.2-mm depth guides give the minimum thickness for the material of 1.5 mm after preparation and finishing.

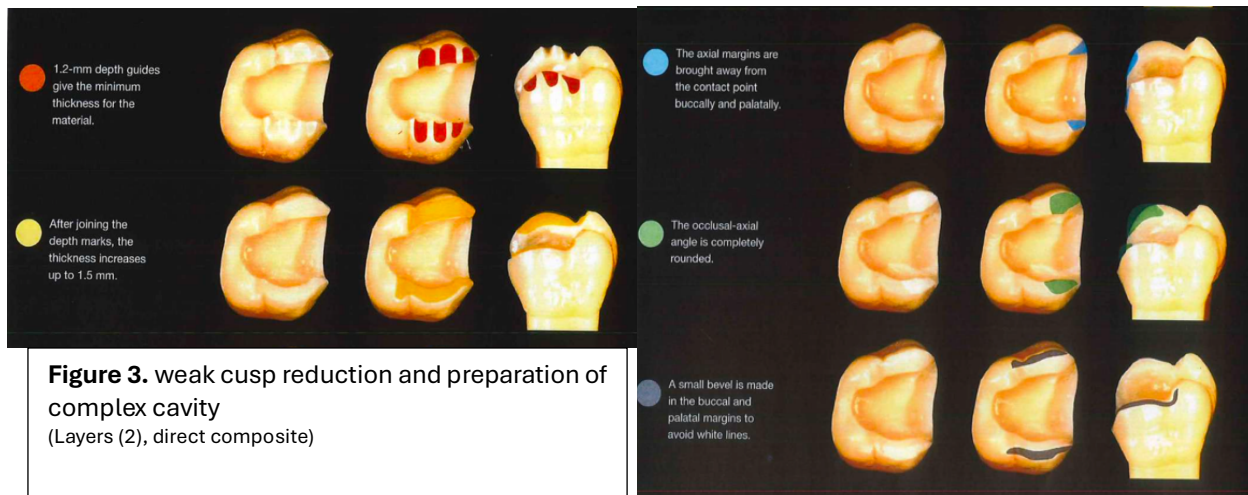
Yellow: After joining the depth marks by bur, the thickness increases around to 1.5 mm.

Blue: The axial margins are brought away from the contact point buccally and palatally.

Green: The occlusal-axial angle is completely rounded. This is to reduce stress concentration area and (resistance form).

Gray: Create a small bevel in the buccal and palatal margins to blend the shade and prevent white lines, but avoid this in high stress areas as the thin composite layer may fracture.

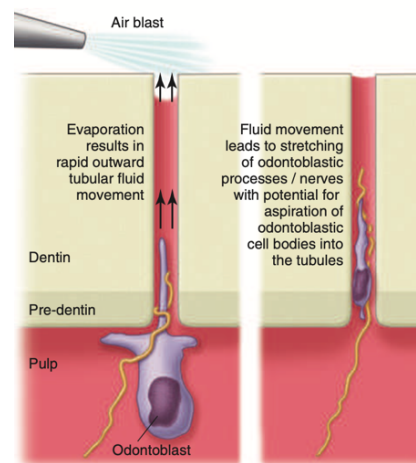
Preparation is designed for medium- and long-term direct bonded restorations. The goal is to carry out all the steps with only one to be able to allow immediate direct restoration with composite resins.



Debridement and Inspection

Debridement (cleaning) of the tooth preparation involves use of the air/water syringe to remove visible debris with water and then excess moisture with a few light bursts of air. Sometimes, debris sticks to walls or corners and must be loosened with an explorer or small cotton pellet. It is important not to dehydrate the tooth by overuse of air as this may damage of the odontoblasts associated with the desiccated tubules (Fig. 4). Complete debridement allows careful inspection of the preparation so as to ensure adherence to all principles of preparation design. Any final changes may then be accomplished, as indicated, followed by steps to disinfect the preparation.

Figure 4: odontoblast changes after air blast.



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Complex Composite Restoration (Restorative Procedure)

Complex cases often involve wide proximal boxes, cusp loss, or deep subgingival. Achieving tight contacts with the adjacent teeth and natural contours is more challenging due to the loss of reference anatomy and increased risk of matrix deformation or open contacts. In some cases, to simplify the procedure, it is recommended to restore the cavity in steps, starting usually with proximal surfaces and transforms the cavity from an open Class II into a contained Class I situation. Once the walls are rebuilt, subsequent composite placement becomes more controlled and predictable.

Isolation:

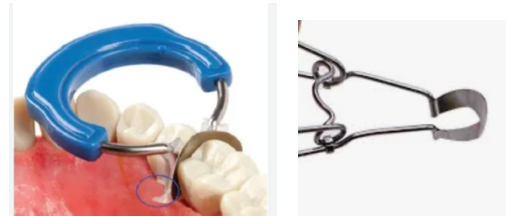
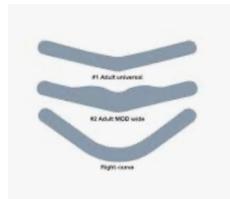
The 1st step of restorative procedure for bonded composite is isolation. Rubber dam is the gold standard for moisture control, visibility, and infection prevention in restorative dentistry. It ensures a dry field, prevents contamination, and improves adhesive performance, leading to more durable restorations. In complex cases, the clinician usually requires to use additional tools and materials such as Teflon, light cured resin gingival barrier, and specific types of clamp e.g. active clamps to hold subgingival tooth structure.

Matrix band selection:

The proximal emergence profile of the restoration and the contact area should be designed to maximize arch continuity and to minimize food impaction. A thorough knowledge of proper contacts and contours and how they ought to be duplicated as well as predisposing factors for proximal contact loss as proximal caries, embrasure size and position, and marginal ridge location and thickness, all of which contribute to achieving ideal occlusal harmony.

Conventional circumferential matrix systems almost end with improper proximal contact points, flat surface, overhangs or open contacts. This is because it cannot produce the contour of the large destructive parts of tooth. However, in some cases with deep margins it can be modified to be used to just elevate the margin and then replaced later with sectional band.

It was found that optimum contact points were highly associated with the sectional matrix system; there was a 70% lower risk for non- optimum contact points than the circumferential matrix system. Sectional matrix systems with separation rings should be implemented as the first choice in clinical decision making for proximal posterior restorations. However, sectional band cannot cover whole the tooth contour as it is small (sections), therefore, the restorative procedure should start with building the proximal walls first using the sectional bands, and then replace the rest of cavity i.e. restore in steps.



Circumferential bands

Sectional bands

Liners:

Deep cavities, are common in complex restorations, require careful management to preserve pulp vitality. The remaining dentin thickness (RDT) determines the need for liners: if RDT is greater than 2 mm, an adhesive system alone is sufficient; if it is between 1 and 2 mm, a resin-modified glass ionomer (RMGIC) liner is recommended; and if it is less than 1 mm, a calcium hydroxide or calcium silicate liner should be applied selectively to the deepest area, covered by GIC.

Cavities with deep margins:

Deep subgingival cavity margins create a series of clinical challenges that make restorative dentistry far more difficult and biologically risky. When the margin drops too far below the gingival level, achieving proper isolation becomes nearly impossible, which leads to contamination from saliva, blood, or crevicular fluid and ultimately compromises bonding quality. Deep margins are also hard to capture accurately during impressions or digital scans because the tissue and fluids obscure the area, increasing the likelihood of distorted or incomplete margins on the final restoration. If a restoration is placed too deep, it may violate the biologic width (2 mm) the protective zone of connective tissue and junctional epithelium that surrounds the tooth. Violating this space triggers a sequence of periodontal problems, including chronic inflammation, attachment loss, pocket formation, gingival recession, and even bone loss. In case of the clinician couldn't get isolated field to apply bonded composite or there is a chance to violate the gingival tissue, crown lengthening or orthodontic extrusion could be the choice to get moisture-free field to apply composite.

It should be noted that dentin or cementum margins have lower bond strength and that it might be difficult to isolate, the open sandwich technique with resin modified glass ionomer cement (RMGIC) may be considered. The open sandwich technique is a restorative dental procedure used primarily for Class II and cervical cavities, especially when the gingival margin of the cavity extends beyond the enamel into dentine or cementum. In this technique; GIC or RMGIC is placed at the base of the cavity, specifically at the gingival margin and axial wall, where it is exposed to the oral environment. Composite resin is then layered above the GIC to restore the remaining cavity and provide optimal aesthetics and strength.

Polymerisation shrinkage:

Polymerisation shrinkage is a significant concern particularly in large cavities, as it can lead to marginal gap formation and postoperative sensitivity and caries initiation. The configuration factor (C-factor), defined as the ratio of bonded to unbonded surfaces, which is higher in Class I and MOD cavities, increasing shrinkage stress. Incremental

placement techniques, such as oblique or horizontal layering in increments of 2 mm or less, and the centripetal build-up technique for Class II cavities, are recommended to reduce polymerisation stress, improve marginal adaptation, and enhance the depth of cure.

Types of composites:

Following adhesive procedure, it is recommended by Sturdevant to use incremental bulk-fill composites as deep dentin-replacement materials because of increased depth of cure (4-5 mm) and reduced shrinkage stress. A conventional nano-hybrid or micro-hybrid composite is recommended for the occlusal surface to replace enamel (1-2 mm) because of superior wear resistance and polishability. A very thin layer of bulk-fill composite might be applied before the adaptation of bulk-fill to fill-up the internal irregularities and reducing void formation.

Occlusal considerations:

Avoid heavy contacts on restoration margins with opposing teeth, adjust occlusion after final curing, and check centric and eccentric contacts. The finishing and polishing sequence should ensure smooth margins, proper anatomical contour, and good proximal contact, using finishing burs, discs and strips, followed by polishing. This is essential for plaque resistance, gingival health, and long-term color stability.

Nevertheless, Indirect restorations should be considered instead of direct composite when there is extensive loss of tooth structure, multiple cusp involvement, poor isolation, or heavy parafunctional habits.
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